

Control Unit Architecture & Hardwired/Microprogrammed Control

Q. What is the role of the Control Unit? Differentiate between Hardwired Control Unit and Microprogrammed Control Unit.

****Definition to Remember****

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the ALU
****Hardw**

1. Hardwired Control Unit

The control logic is implemented physically using digital hardware circuits (logic gates, flip-flops, decoders).

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2. Microprogrammed (Soft-Wired) Control Unit

The control signals are generated by executing a tiny, low-level program (a Microprogram) stored in a special, fast memory called Control Memory (ROM).

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3. Key Differences

| Feature | Hardwired CU | Microprogrammed CU |

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****Must-Write Points (for 10 marks)****

1. ****Cor**

*****Quick Recall*****

`Control
Slower,